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Benchmarking Provably Safe Reinforcement Learning

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Benchmarking Provably Safe Reinforcement Learning

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***Abstract*—Reinforcement Learning (RL) has found applications in various safety-critical environments, where ensuring safety is paramount. In this work, we benchmark different provably safe RL methods, focusing on continuous action masking to restrict the agent's actions to a safe set. This safe action set is represented using zonotopes, which allow the agent to avoid unsafe states during training. We extend existing safe RL methods to diverse Mujoco environments, including Half-Cheetah, Ant, Walker2D, and Humanoid, and analyze their learning behavior. Our experiments evaluate the impact of action set dimensionality on training effectiveness and investigate the influence of different zonotope shape approximations (L1 vs. L2). We also explore the performance of safe RL methods in both state-independent and state-dependent environments, using ray action masking, action replacement, and Proximal Policy Optimization (PPO). The results show that action masking significantly improves convergence rates and sample efficiency. However, we identify challenges in high-dimensional action spaces, where the safe action set becomes increasingly restrictive. Future work aims to address these challenges, particularly the high-dimensional problems with L1/L2 approximating zonotopes, and extend provably safe RL methods to off-policy algorithms.**

I. Introduction [LS]

Artificial intelligence is integrated into our lives at a rapid pace and Reinforcement Learning (RL) is already used in many real world tasks such as traffic signal control [1], finance [2] or energy management [3]. And as RL is used more in the real world, the question of safety is also ever more important. If RL is to be deployed in environments where its decisions could harm humans, we need to guarantee that it does not do so. We call these safety critical environments and aim to achieve provable safety.

Hans et al. [4] defines an agent as safe, if the agent never reaches an unsafe state. In order to never reach an unsafe state, the agent must only perform safe actions. We consider these safe actions as part of the safe action set, which is a subset of the entire continuous action set of our RL agent. We represent this safe action set with a zonotope, which is a convex geometric structure. For the purposes of these experiments we assume that the safe action set is computable and representable by a zonotope.

In this work we benchmark different methods of provably safe RL, which were adapted from Stolz et al. [5], a paper which proposes three different methods of continuous action masking as a way to focus learning to a relevant action set. This was done in order to increase training effectiveness and enable the use of RL in safety critical environments. We aim to extend this work by:

1) Extending provably safe RL methods to diverse Mujoco environments [6] with varying objectives (half-cheetah, ant, walker2D, and humanoid) and analyzing their learning behavior.
2) Implementing safe input set calculators for three different quadrotor environments in both 2D and 3D using zonotope representations for state and action sets, and assessing how action dimensionality affects numerical convergence rates.
3) Investigating the behavior of the provably safe learning algorithms across different zonotope shape approximations (L1 vs L2) and in high-dimensional action spaces.

Of the methods proposed by Stolz et al. [5] ray action masking was chosen for our experiments, due to it consistently achieving the best training results. Both ray action masking and action replacement use the convex safe action sets. Both are evaluated on different environments using proximal policy optimization (PPO) compared to a baseline without either of the two methods.

We implement new mujoco environments from the gymnasium library[6], which have state independent action sets. In the mujoco environments we analyze the impact of an increasing amount of dimensions in the action set on action masking and action replacement. Furthermore we explore the impact of the safe action set shape, by comparing L1 and L2 zonotopes. We also implement and benchmark on different quadrotor environments, which are state dependent.

II. Relevant literature [MAM]

The concept of safe reinforcement learning was first formally distinguished by Garcia and Fernandez [7], establishing a foundation for subsequent research in this field.

Recent work in provably safe reinforcement learning can be categorized into several approaches across diverse applications, as conceptualized by Krasowski et al. [8], who provided a consistent framework for both discrete and continuous action spaces. Building on this foundation, Kochdumper et al. [9] applied reachability analysis-based methods using polynomial zonotopes to autonomous driving and quadrotor systems.

Several domain-specific implementations have emerged. For aerial vehicles, Shao et al. [10] developed safety-critical approaches. In autonomous driving, Fulton et al. [11] explored provably safe methods. Ceusters et al. [12] applied these techniques to power systems with known constraints in multi-energy management systems. In human-robot interaction scenarios, Thumm et al. [13] advanced safe reinforcement

learning for robotic manipulation. Extending beyond physical systems, Seetanadi et al. [14] applied safe RL methods to improve communication across computer networks.

On the algorithmic front, Wabersich et al. [15] developed a safety filter for data-driven control applications. Huang et al. [16] demonstrated how masking irrelevant actions can improve training efficiency in environments with high-dimensional action spaces. Building on this concept, Hou et al. [17] compared invalid action masking with off-policy and on-policy algorithms in real-time strategy games. Further advancing this line of research, Stolz et al. [5] showed how provably safe action masking methods can enhance sample efficiency and improve convergence.

III. METHODOLOGY [LS]

Reinforcement learning with policy gradients is a recent and effective model free sequential decision making. This section explains reinforcement learning with policy gradients. And how provably safe methods are build on top of vanilla PG. Zonotope representations. Important definitions.

A. Reinforcement learning and proximal policy optimization

Every method used in our experiments use proximal policy optimization (PPO) [18], a popular RL algorithm which improved upon existing policy gradient methods. Policy gradient methods aim to improve the expected reward, by using an estimator off the policy gradient and updating the policy parameters using a gradient ascent algorithm. The policy gradient is

$$\nabla J(\pi_\theta) = E_{\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) A_{\pi_\theta}(a,s)\right], \tag{1}$$

with $A_{\pi_\theta}(a,s)$ being the advantage function, which calculates the expected improvement in reward by taking an action a in the state s, over the average action in this state selected by the policy $\pi_\theta(a|s)$. PPO improves upon these methods by introducing a clipped surrogate objective function with restricts excessively large policy updates.

B. Safe Action Set and Zonotope construction [YMU]

In the context of reinforcement learning, continuous action spaces can be defined as multidimensional intervals. These spaces can become quite large, especially with a higher number of action dimensions. Each action from this action set A could result in the agent reaching either a safe or unsafe state. If we are able to determine a convex subset of actions that will result in the agent reaching safe states, we can label these actions as the safe action space A_s and represent it with a zonotope.

A zonotope is a convex polytope that is the Minkowski sum of a finite set of line segments. In other words, a zonotope can be represented as the set of all points that can be reached by adding scaled versions of a set of generator vectors [19]. Formally, a zonotope Z centered at point p, with generator vectors $v_1, v_2, \ldots, v_n$, is given by:

$$Z = \left\{ p + \sum_{i=1}^{n} \delta_i v_i \mid -1 \leq \delta_i \leq 1 \right\}$$

Where:

- p is the center of the zonotope,
- v_i are the generator vectors, which define the directions in which we can move within the zonotope,
- δ_i are scaling factors within the interval $[-1, 1]$, allowing movement along each generator vector [19].

Thus, the zonotope represents a convex region defined by the sum of these line segments. In the context of safe action spaces in reinforcement learning, the zonotope allows us to efficiently represent a region of safe actions, where any action within this space will result in the agent reaching a safe state.

C. Action Replacement

Action replacement can be used to ensure provable safety. It does so by replacing possibly unsafe actions returned by the agent with a safe action.

The policy samples an action a from the action space A, by checking if the action part of the convex safe action set A_s we can determine if the action is provably safe or possibly unsafe. If a possibly unsafe action was chosen, a new action is sampled from the safe action set. This action can either be randomly sampled or a predefined fail-safe action. [9] While this method can ensure provable safety, it is possible that action replacement can impede the learning process of the agent. If an unsafe action is randomly replaced by a safe action, the performed action an resulting reward have no connection to the action originally selected by the policy. This may lead to the replacement agent having difficulty in learning from its mistakes.

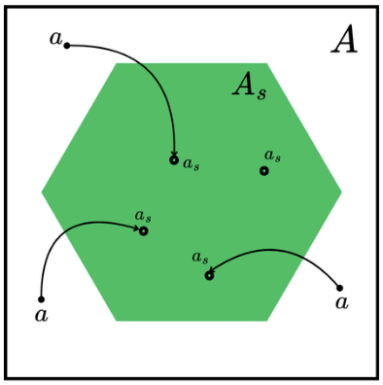


Fig. 1. Action replacement. If an action returned by the agent is unsafe, replace it by a randomly sampled action from the safe action set. Figure adapted from [5]

D. Ray Action Masking

Ray action masking[5] utilizes the convex structure of the safe action zonotope in order to map the entire action space A to the safe action space A_s. It achieves this by scaling each action in A alongside a ray from the center of A_s to the boundary of A. By using the function $g_s(a)$ to map each action a sampled by the policy $\pi_\theta(a|s)$, a safe policy $\pi^s{}_\theta$ is created.

$$a^s = g_s(a) = c + \frac{\lambda_{A^s}(a)}{\lambda_A(a)}(a - c) \tag{2}$$

With $\lambda_A(a)$and $\lambda_{A^s}(a)$ representing the distances of an action a to the boundaries of the safe action set and action space. Both measured from the center of the safe action set zonotope c in the direction of a. This new policy can only sample safe actions a_s. Using this function, the updated policy gradient is:

$$\nabla_\theta J(\pi^s{}_\theta(a^s|s)) = E_{\pi^s{}_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) A_{\pi^s{}_\theta}(a^s, s)\right] \quad (3)$$

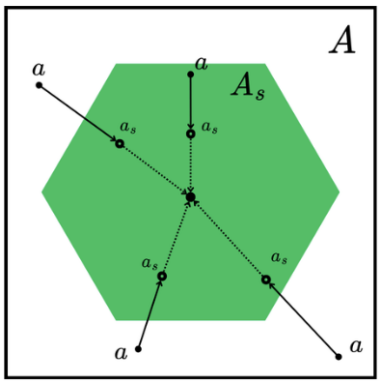


Fig. 2. Ray action masking. Map all actions along a ray pointing towards the center of the zonotope. All actions will get mapped to a new value. Figure adapted from [5]

IV. Implementation

The state set, safe state set, continuous action and safe action sets are represented using Zonotopes. We implement the code base in python using the gymnasium [6] library for environment simulation and tianshou [20] library for running the proximal policy optimization. We distinguish the implementation on two types of action sets. Namely, state dependent and independent action sets.

A. State independant action sets [YMU]

State-independent action sets describe environments that lack a concept of an absolute safe state, meaning the action space remains unchanged regardless of the state. However, action masking and action replacement can still be applied by defining a constant relevant action set that is independent of any given state. For consistency, we will refer to this set as the "safe set" throughout the paper.

We conducted our state-independent experiments in Mujoco environments, where multi-joint robots are controlled by actuator motors to learn and perform tasks such as stabilization and walking [6]. In this context, the safe action set is defined by constraining the cumulative power output of all joints.

Two types of experiments were performed in the Mujoco environments. First, we evaluated the impact of the initial safe action set on performance by comparing L1 and L2 norm-based zonotopes in the Mujoco Walker2D environment [21]. In the second experiment, we compared action masking and action replacement with the PPO baseline model across two different Mujoco environments.

Analysis of L1 vs L2 zonotope

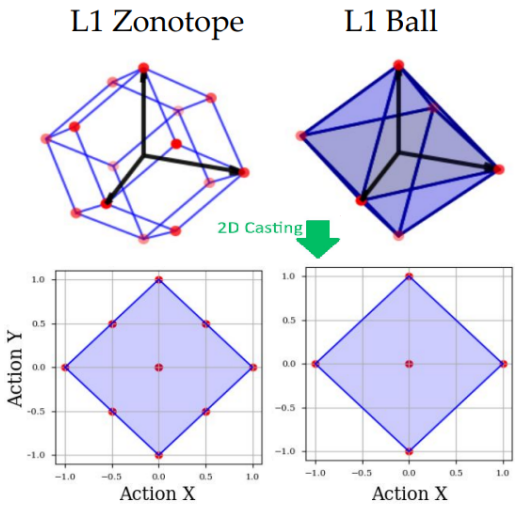


Fig. 3. Hypothetical L1 norm on the right and L1 Zonotope on the left.

We compared the performance of L1 and L2 norm-based zonotopes in the Walker2D environment, a bipedal robot simulation where the goal is to move forward by applying torque to six joints. The action space consists of continuous torques applied to the six joints of the robot, while the observation space includes the robot's positions, velocities, and joint angles. The objective is for the robot to walk in the forward direction by appropriately controlling these joints to maintain balance and propulsion [21].

The L1 and L2 norms were not perfectly represented by zonotopes. Instead, we constructed zonotopes based on L1 and L2 norms properties.

1) L1 Zonotope Implementation: As it was stated above, in order to make action masking and action replacement relevant to the Walker2D environment, we added an additional constraint by limiting the cumulative power output based on the L1-norm. With this constraint, our static safe action set can be described as:

$$\mathcal{A}^s = \{a \mid \|a\|_1 \leq \alpha_p\} \quad (4)$$

Where a_p represents the maximum total power output which was set as 1.

The L1 norm cannot be represented in three dimensions or higher, as demonstrated in Figure 3, where the hypothetical L1 norm ball features triangular faces that cannot be represented by a zonotope, which only has parallelogram faces [19]. Therefore, we used a zonotope that overestimates the L1 norm while preserving its properties between any two actions. In other words, our zonotope can describe the static safe action set as:

$$\mathcal{A}^s = \{a_i, a_j \mid \|a_i\| + \|a_j\| \leq \alpha_p\} \quad (5)$$

Where a_i and a_j describe any two actions in the environment. To generate this zonotope, we require every combination of generator vectors consisting of 1 and -1. We can deduce the

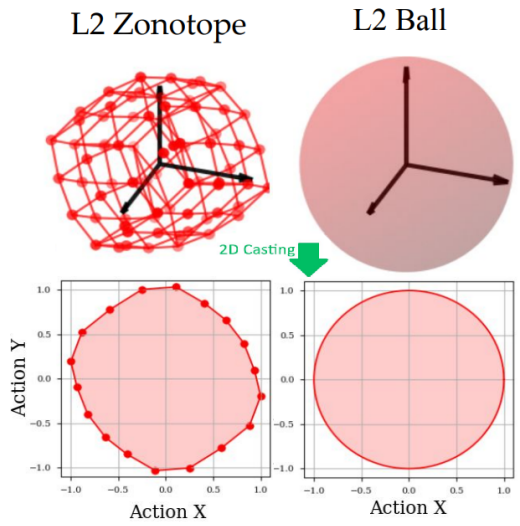


Fig. 4. Hypothetical L2 norm on the right and L2 Zonotope on the left.

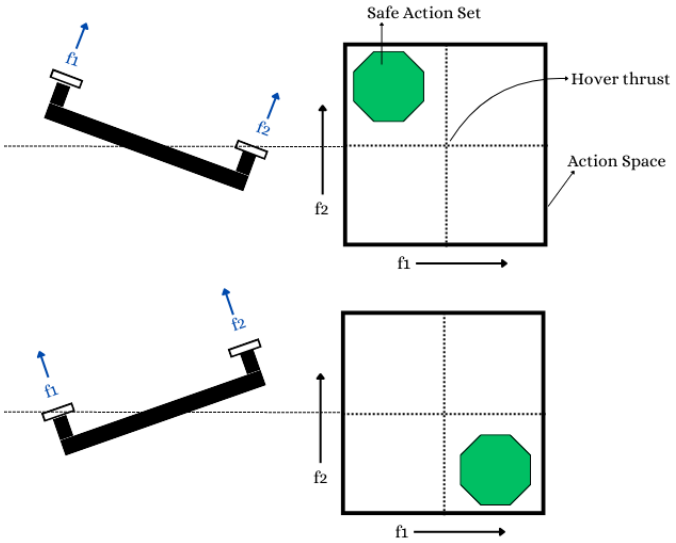


Fig. 5. Visualisation of state dependent set of the quadrotor 2-D with coupled dynamics. The box on the right represents space of possible actions. The green zonotope represents safe action set. When the quadrotor is tilted to the clockwise (top half), the safe action set adjusts into the top left quadrant where $f_2 > f_1$ and allowed actions guide the quadrotor back to its hover position. Vice versa with the bottom half.

number of generators by removing symmetrical generator vectors, which results in $2^{(\text{dimension}-1)}$ vectors. Since the Walker2D environment has 6 dimensions, we need 32 generator vectors to generate our L1 Zonotope.

2) L2 Zonotope Implementation: Similar to the L1 norm, we can also set the constraint on cumulative power output based on the L2 norm. With this constraint, the static safe action set can be described as:

$$\mathcal{A}^s = \{a \mid \|a\|_2 \leq \alpha_p\} \tag{6}$$

Similar to the L1 norm, the L2 norm is not a convex shape and therefore cannot be described by a zonotope [19]. To address this, we constructed the L2 Zonotope by sampling generator vectors using the Gram-Schmidt process [22]. We found that using $dimension^2$ generator vectors is sufficient to approximate the L2 norm, as shown in Figure 4. In the Walker2D environment, this corresponds to 36 generator vectors required for constructing the L2 Zonotope.

Action Masking and Action Replacement in Mujoco Environments

We compared action masking and action replacement in the Mujoco Ant and Mujoco Half-Cheetah environments, using PPO as our baseline model.

The Ant environment is a 3D quadruped robot simulation where the goal is to move forward by applying torque to eight joints connecting the torso and legs. The action space consists of continuous torques applied to these joints, and the objective is for the robot to coordinate its legs to walk forward while maintaining balance [23].

The Half-Cheetah environment is a 2D robot simulation where the goal is to apply torque to six joints to make the robot run forward as fast as possible. The action space includes continuous torques applied to the robot's joints, and the objective is to maximize forward movement while avoiding backward motion [24].

As discussed earlier, we introduce an additional constraint on the total energy output of the joints to enable action masking and action replacement in Mujoco environments. For both environments, we tested only the L2 Zonotope to describe this constraint. This decision was due to the Mujoco Ant environment having 8 dimensions in its action space, which requires 128 generator vectors for constructing an L1 Zonotope, compared to just 64 for an L2 Zonotope. Conducting experiments with the L1 Zonotope was computationally expensive, so we decided to limit our tests to the L2 Zonotope.

B. State dependent action sets [MAM]

State dependent action sets need to be computed for each time step. The objective is to maximize the volume of the safe action set zonotope. Here we maximize the approximation of the volume over generator scaling vectors and center of the safe action zonotope as described by Stolz et al. [5]. The online quadratic optimization problem is described as

$$\max_{p} \quad \hat{\text{Vol}}(A_s(p)) \tag{7}$$

where $\hat{\text{Vol}}$ is an approximation to the volume of the zonotope and $p = [c, \tilde{p}]$ with c being the center of zonotope and $\tilde{p}$ being the independent scaling factors for the generators of the safe action set zonotope A_s calculated for each discrete time step. The constraints of the problem 7 are detailed in Stolz et al. [5, Eq. (33)]. The constraints ensure that the next state by following any action from A_s leads to a verified safe state.

We implement the action masking and action replacement for 3 quadrotor environments. These include adapted versions

of 2-D quadrotor [25], a 2-D quadrotor with coupled dynamics [25] and a 3-D quadrotor [26].

The non-linear dynamical system of quadrotors is linearized around a hover point and can be represented in terms of matrices A, B and E as:

$$s^{t+1} = As^t + Ba^t + Ew^t$$

Where s^t is the state at time t, a^t is the action at time t and w^t is a bounded disturbance uniformly sampled at time t.

1) Quadrotor 2-D with coupled dynamics: Quadrotor 2-D [25] with coupled dynamics has the state $s = [x, z, \dot{x}, \dot{z}, \theta, \dot{\theta}]^T$, the inputs $a = [f_1, f_2]^T$ describe independant motor thrusts and the disturbance $\mathcal{W} = [w_1, w_2] = [-0.08, -0.08] \times [0.08, 0.08]$ applied to dx and dz. The systems dynamics are defined as

$$\dot{s} = \begin{bmatrix} \dot{x} \\ \dot{z} \\ (f_1 + f_2)k\sin(\theta) + w_1 \\ g + (f_1 + f_2)k\cos(\theta) + w_2 \\ \dot{\theta} \\ -d_0\theta - d_1\dot{\theta} + n_0(-f_1 + f_2) \end{bmatrix}$$

Where $g = 9.81ms^{-2}$, $k = 0.636$/kg, $d_0 = 70$, $d_1 = 17$ and $n_0 = 25$. The dynamics are linearized at the hover point $s = [0, 1, 0, 0, 0, 0]$, which is also the goal state s^* using taylor approximation to obtain the linearized form. The system matrices A and B are adapted from [25] for independent motor thrust actions. The reward function for action a and state s penalizes deviation from the goal state and is given by

$$r(a, s) = \exp\left(-\|s - s^*\|_2 + -\frac{0.01}{2}\left\|\frac{a - a_{\min}}{a_{\text{range}}}\right\|_1\right) \quad (8)$$

Where $a_{\min}$ is the vector of lower bound of actions over all dimensions and a_{range} is the absolute difference of upper and lower bound actions. Action space is a box in 2-D space represented as a zonotope and its lower and upper bounds are $[2.71, 2.71]$ and $[12.71, 12.71]$ respectively. State space constraints are a box represented as a zonotope in 6-D space and their lower and upper bounds are $[-1.7, 0.3, -0.4, -0.5, -\pi/12, -\pi/2]$ and $[1.7, 2.0, 0.4, 0.5, \pi/12, \pi/2]$ respectively. The zonotope template matrix is given by $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{bmatrix}$

2) Quadrotor 2-D: Quadrotor 2-D has the same state and disturbance as the one with coupled dynamics with a different input space $a = [F, \mathcal{T}]$ where F and $\mathcal{T}$ are cumulative force and torque on the quadrotor. The dynamics are defined as

$$\dot{s} = \begin{bmatrix} \dot{x} \\ \dot{z} \\ Fk\sin(\theta) + w_1 \\ g + Fk\cos(\theta) + w_2 \\ \dot{\theta} \\ -d_0\theta - d_1\dot{\theta} + n_0\mathcal{T} \end{bmatrix}$$

The constants k,g,n_0,d_0, d_1, linearization point, goal state and reward function are the same as quadrotor 2-D with coupled dynamics. The system matrices A and B are taken from [25]. Action space is represented similar to Quadrotor 2-D with lower and upper bounds $[13.931, -0.261]$ and $[16.931, 0.261]$ respectively. The state constraints are same as Quadrotor 2-D with coupled dynamics. The generator template matrix of the safe action set zonotope A_s is given by an identity matrix of dimension 2.

3) Quadrotor 3-D: Quadrotor 3-D has a 12 dimensional state space $s = [x, y, z, \dot{x}, \dot{y}, \dot{z}, \phi, \theta, \psi, \dot{\phi}, \dot{\theta}, \dot{\psi}]$. The disturbance set, given as $\mathcal{W} = [w_1, w_2, w_3] = [-0.08, -0.08, -0.08] \times [0.08, 0.08, 0.08]$ is applied to $\dot{dx}$, $\dot{dy}$ and $\dot{dz}$ during state transition. The inputs are $a = [F, \ddot{\phi}, \ddot{\theta}, \ddot{\psi}]$ where F is the cumulative force on the quadrotor, $\ddot{\phi}$, $\ddot{\theta}$ and $\ddot{\psi}$ are the angular accelerations of three angles for roll - ϕ, pitch - θ and yaw - ψ. The differential equations of the system are normalized so that the system is independent of mass. The system can be described by

$$\dot{s} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ -F\cos(\phi)\sin(\theta) + w_1 \\ F\sin(\phi) + w_2 \\ -g + F\cos(\phi)\cos(\theta) + w_3 \\ \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \\ \ddot{\phi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix}$$

The hover point for linearization is $[0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]$ and the system matrices A and B are given by Kaynama et al. [26]. The reward function is same as 8. Action space is a box in 4-D space represented as a zonotope and its lower and upper bounds are $[-14.81, -0.5, -0.5, -0.5]$ and $[7.81, 0.5, 0.5, 0.5]$ respectively. State space constraints are a box represented as a zonotope in 12-D space and their lower and upper bounds are $[-1.5, -1.5, -0.5, -1.5, -1.5, -1.5, -\pi/12, -\pi/12, -\pi/6, -1, -1, -1]$ and $[1.5, 1.5, 2.5, 1.5, 1.5, 1.5, \pi/12, \pi/12, \pi/6, 1, 1, 1]$ respectively. The generator template matrix of the safe action set zonotope A_s is given by an identity matrix of dimension 4.

V. Experimental Results [Shared]

Below we describe the experimental set up and discuss numerical results.

A. Quadrotor Experiments [MAM]

The figure 6 shows average reward per step throughout the training process. We trained our model 5 times with different initial seeds and trained upto a total number of 150.000 steps. The models used a 2-layer neural network. Both of the 2-D quadrotors see a significant improvement in numerical convergence rate of ray action masking over

baseline (PPO) and action replacement. It can be observed that, by changing the action dimensions from quadrotor 2-D to quadrotor 2-D with coupled dynamics, action replacement sees an improvement over baseline despite environment physics being the same. This shows that changing how the action dimensions are set up, can impact how the learning algorithm behaves. In the experiments for quadrotor 3-D, we see no improvement of action masking or action replacement over baseline.

B. L1 and L2 Zonotope Comparison[YMU]

Performance Results

We conducted our experiment in the Walker2D environment with an additional energy consumption constraint. For both L1 and L2 static safe action space definitions, we trained our model five times for 10^6 steps. In Figure 7, we compare the average reward curves for our models, with transparent bootstrapped 95% confidence intervals. As shown in the graph, we did not observe a significant performance difference between the initial zonotope configurations. However, the confidence interval for the L1 configuration increases in the later stages of training compared to the L2 configuration. While the performance does not change drastically on average, the L2 configuration appears to yield more stable results. Due to time and computational limitations, we were unable to test other zonotope initializations and observe their effects, which could be explored in future work.

Efficiency Results

Although we did not observe any performance difference between L1 and L2 initialization, both zonotopes may still have their uses. First, we analyzed how our zonotopes overestimate or underestimate the hypothetical L1 and L2 balls. To do this, we computed the hypothetical L1 volume using:

$$V_{L1} = \frac{(2r)^n}{n!} \tag{9}$$

And L2 volume with:

$$V_{L2} = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} r^n \tag{10}$$

For the volumes of L1 and L2 zonotopes, we used the center and generator vectors corresponding to each zonotope [19]. Due to the sampling process, the L2 norm results in slightly different zonotopes, though the volume does not change significantly. As shown in Figure 8, we observe that the hypothetical L1 volume is overestimated, while the hypothetical L2 volume is underestimated.

It is important to note that the L1 configuration is not suitable for environments with large action spaces. As seen in the graph, the action volume approaches zero as the action dimension increases. Additionally, the number of generator vectors required to construct an L1 Zonotope grows exponentially, whereas for the L2 Zonotope, it increases quadratically. Since our calculations rely on the number of generator vectors, training times follow this growth pattern, making the L1 Zonotope less practical for environments with larger action spaces.

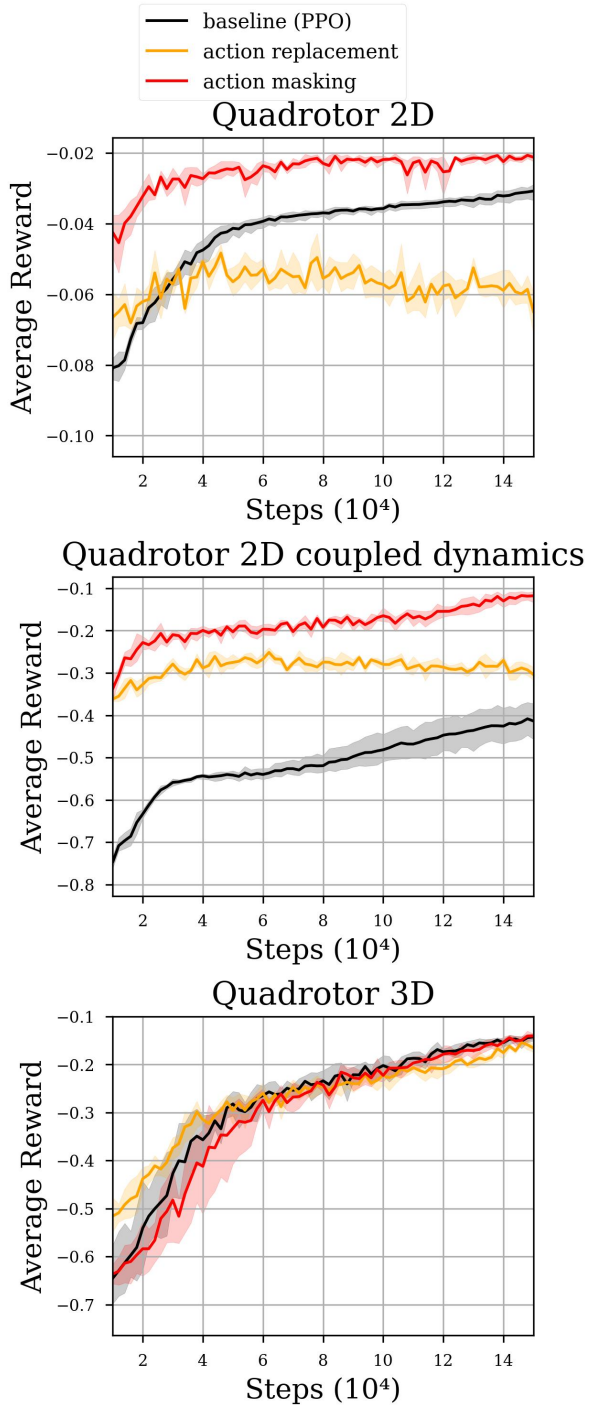


Fig. 6. Average reward per step computed over each collection throughout the training process.

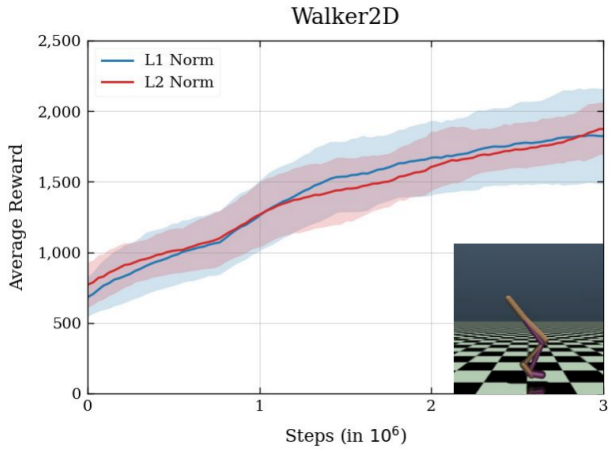


Fig. 7. L1 and L2 Zonotope comparision on Walker2D enviroment with average reward curves with transparent bootstrapped 95% confidence interval.

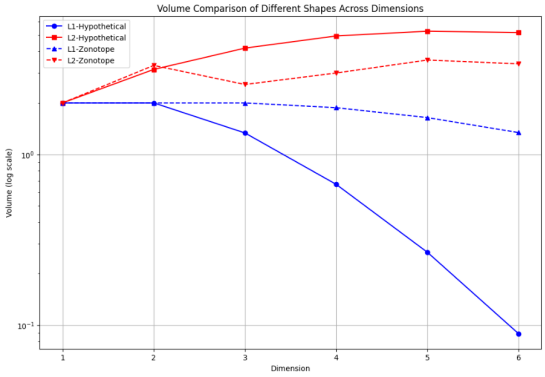


Fig. 8. Volume Comparision of Hypothetical and Zonotope interpretation of L1 and L2 norms with respect to action space size.

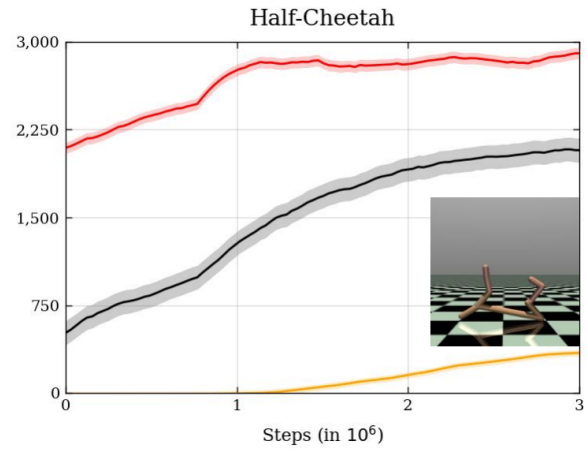


Fig. 9. Action Masking and Action Replacement in Mujoco Half-Cheetah environment average reward curves with transparent bootstrapped 95% confidence interval.

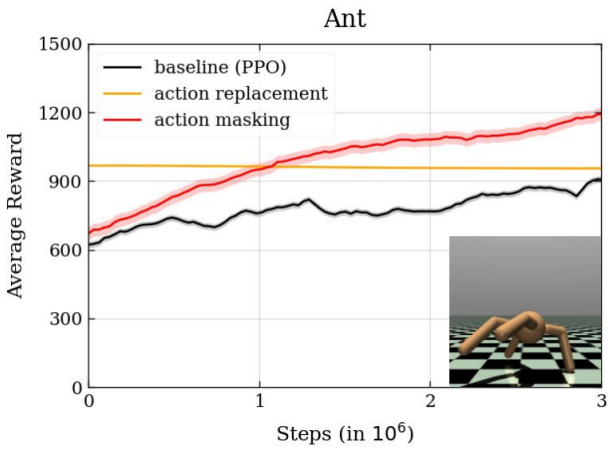


Fig. 10. Action Masking and Action Replacement in Mujoco Ant environment average reward curves with transparent bootstrapped 95% confidence interval.

While the L1 Zonotope is computationally more efficient for environments with action spaces of six dimensions or fewer, we chose to conduct further experiments using only the L2 Zonotope, as all the environments we tested had action spaces of six dimensions or more.

C. Mujoco Environments [LS]

Experiments run in the half cheetah environment show that ray action masking clearly outperforms both the baseline and action replacement agents, see Figure 9. Using ray action masking average rewards of nearly 3000 were achieved. This mirrors not only the results achieved in Stolz et al. [5], but also our own experiments. It is also notable that the replacement agent achieved the lowest rewards, even lower than the baseline implementation. The graph also show that in the first 10^6 of a total $3*10^6$ steps, no significant improvement of the agents performance is registered. The half cheetah environment calculates reward as forward movement minus the control cost. Since the control cost is restricted by the safe action set, we can assume that the replacement agent had difficulty in learning actions which result in forward motion.

In comparison to the six action dimensions of half cheetah and walker2D environments, the ant environment has eight. This comparative increase in action dimensions is significant, as the resulting safe action set is more restrictive than in lower dimensions. Our experiments in the ant environment show that ray action masking achieved average rewards of 1200 after training, which is again higher than both the baseline and replacement agents, as seen in Figure 10. It is notable that the replacement agent returned a constant reward of close to 1000 in every step. This suggests that the agent was not able to learn anything in this environment. The cause of this could be the increase in action dimensions. If the replacement agent were to only sample actions outside its safe action space, each of these actions would be replaced. This replacement of every action could be responsible for the agents inability to learn.

VI. Conclusion [MAM]

We showed how provably safe RL methods action replacement and ray action masking work on diverse set of environments including multiple quadrotor environments rep-

resenting safety constraints and multiple Mujoco environments representing relevant RL. The methods proposed work in continuous action space, discrete time and use reachability analysis to ensure transition to a verified safe state. The numerical experiments show that action masking can significantly improve the convergence rate and sample efficiency for the PPO algorithm. However, there are some caveats. First, there is a significant computational overhead cost in particular for state dependent environments which require solving an online quadratic optimization problem and for high dimensional Mujoco environments. Secondly, the performance of action masking depends on how the action dimensions are chosen. Decoupling the action dimensions (or coupling the system dynamics) leads to a better performance of action masking and action replacement over baseline. Additionally, the high dimensions L1/L2 approximating zonotopes become increasingly restrictive in high dimensions. Which means the relative volume of safe action zonotope becomes prohibitively small compared to the total area of the feasible action space. Finally, there is considerable domain input required to gain effective training and benefits from safe RL methods.

Future research directions. Moving forward, we would like to see a solution to the high-dimensional problem for L1/L2 approximating zonotopes. We tried to run experiments of Humanoid environment [6] which has a 17 dimensional action space. The agent could not learn anything significant because of the restrictive safe action set zonotope. We would also like to see the numerical results of Quadrotor 3-D environment with coupled dynamics. That would mean to have the action dimensions as independent motor thrusts. Additionally, the safe action set calculator only deals with zonotopes with linear boundaries and continuous sets which can be restrictive. An interesting research direction is to extend this method to discontinuous or non-convex sets. Work by Kochdumper et al. [9] is an interesting extension in this direction using parametrized reachability analysis and polynomial zonotopes. Finally, off-policy algorithms like SAC [27] and Twin Delayed DDPG [28] are popular in applications. Extension of provably safe methods to these algorithms is also an interesting research direction.

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